

Name: _____



Checking and securing dividing a fraction by a whole number

Task A: Contexts for dividing fractions

- 1) A pizza is split into 12 equal pieces. A teacher takes 2 pieces and the remaining is given equally between 2 students. What fraction of the whole pizza does each student receive?

- 2) There are 14 crayons in a box. Two are lost and the rest are shared equally between a table of 4 students. What fraction of the full crayon box does each student receive?

- 3) A chocolate bar has 24 equal pieces. The teacher eats 6 pieces and shares the rest between 3 students. The teacher says, everyone has had the same fraction of the chocolate bar. Is the teacher correct?

- 4) A pizza is cut into 4 equal pieces. There are 6 people who need to share the pizza equally. How should the pizza be cut so everyone receives an equal piece? What fraction does each person receive?

Task B: Dividing any fraction by a whole number

- 1) Match the questions and answers.

$$\frac{12}{19} \div 4$$

$$\frac{4}{19}$$

$$\frac{12}{19} \div 3$$

$$\frac{3}{19}$$

$$\frac{12}{19} \div 2$$

$$\frac{6}{19}$$

2) Work out the answers to the following. You can draw bar models if needed.

a) $\frac{24}{25} \div 6 =$

b) $\frac{21}{31} \div 7 =$

c) $\frac{9}{11} \div 3 =$

d) $\frac{10}{19} \div 5 =$

3) Jacob has used equivalent fractions to work out his answer but has spilt ink all over his work. Can you identify what his missing numbers were?

a) $\frac{3}{2} \div 2 = \frac{\text{[ink]}^{\text{[ink]}}}{2 \times 2} \div 2$
 $= \frac{\text{[ink]}^{\text{[ink]}}}{\text{[ink]}^{\text{[ink]}}} \div 2$
 $= \frac{3}{4}$

b) $\frac{5}{8} \div 4 = \frac{\text{[ink]}^{\text{[ink]}}}{\text{[ink]}^{\text{[ink]}}} \div 4$
 $= \frac{20}{32} \div 4$
 $= \frac{\text{[ink]}^{\text{[ink]}}}{\text{[ink]}^{\text{[ink]}}}$

4) Work out the answers to the following, giving your answer in its simplest form.

a) $\frac{5}{9} \div 4 =$

b) $\frac{6}{11} \div 5 =$

c) $\frac{10}{13} \div 4 =$